DO GIA HUY

ML/AI Engineer Intern

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CAREER OBJECTIVE

AI undergraduate at Swinburne University of Technology, specializing in Generative AI, RAG systems, and deep learning. Passionate about building end-to-end AI solutions that connect research to production. Actively expanding expertise in MLOps and cloud deployment to deliver scalable, real-world AI applications.

EDUCATION

Swinburne University of Technology — **Expected Graduation: 2027**

- Bachelor of Artificial Intelligence
- GPA: 3.43 / 4
- Relevant Coursework: Artificial Intelligence, Intelligent Systems, Cloud Architecture, and Database Design

SKILLS

- **Programming Languages & Frameworks:** Python (LangChain, Chroma, TensorFlow, Scikit-learn), C#, JavaScript, Node.js, RESTful APIs, JSON
- **AI & Machine Learning:** ML, DL, Generative AI, RAG, Prompt Engineering
- **Languages:** English – PTE 62 (~IELTS 6.5) (2024)
- **Databases:** ChromaDB, PostgreSQL (Supabase)
- **Tools:** Git, AWS (EC2, S3, Lambda), Docker (familiar), Google Colab
- **Soft Skills:** Teamwork, Self-study, Problem solving, Critical thinking

PROJECTS

LLM-BASED AUTONOMOUS AGENTS FOR CYBERSECURITY (5 Members | Role: Project Coordinator & LLM Developer)

- **Description:** A locally deployed multi-agent system for cybersecurity threat analysis and incident response, powered by RAG and LLMs.
  - Designed and owned the full RAG pipeline, grounding LLM outputs in CTI sources using LangChain and Chroma.
  - Engineered the document ingestion pipeline, includes parsing, chunking, and embedding knowledge bases into a vector store with sub-2s retrieval latency.
  - Applied prompt engineering and retrieval grounding to achieve ~30% improvement in contextual relevance over baseline LLM outputs.
- **Tools:** Python, LangChain, Chroma, OpenRouter API, OpenAI API, Git
- **Github:** https://github.com/nghiatran0401/cyber-llm-agent/tree/cos30018

TRAFFIC-BASED ROUTE GUIDANCE SYSTEM (4 Members | Role: Team Leader & AI Developer)

- **Description:** A machine and deep learning system for predicting traffic conditions and recommending optimal routes across a real road network in Boroondara.
  - Architected a predictive model combines LSTM & GRU networks with graph-based routing logic to minimize travel time.
  - Built a preprocess pipeline to handle ~400,000 time-series traffic records, resolving data leakage and structural issues.
  - Benchmarked LSTM & GRU models, achieving ~82% prediction accuracy (18% MAPE), outperforming the baseline by 2%.
  - Managed team collaboration and model versioning using Git with structured branching across all project phases.
- **Tools:** Python, TensorFlow, Scikit-Learn, Pandas, Matplotlib, NumPy, Git
- **Github:** https://github.com/COS30019-IntroductiontoAI/Traffic-based-Route-Guidance-System
- **Link:** https://cos30019-introductiontoai.github.io/Traffic-based-Route-Guidance-System/route-guidance

SMART HOME MANAGEMENT SYSTEM (4 Members | Role: Full-Stack Developer & Database Designer)

- **Description:** A full-stack web application for managing IoT smart home devices, including monitoring, scheduling, and real-time control.
  - Designed a relational database schema in PostgreSQL to support IoT device management, scheduling, and state persistence.
  - Built RESTful API endpoints and backend services using Node.js for device data monitoring, control, and alert handling.
  - Developed comprehensive dashboard interfaces for real-time device status tracking and event-triggered notifications.
- **Tools:** React, Node.js, PostgreSQL, REST APIs, Git
- **Github:** https://github.com/Robertise/Smart-Home-Management-System